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Probable realization of rotor systems in $SrTiO_3$ and $PbZr_{1-x}Ti_xO_3$

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ABSTRACT

Rotor descriptions of some perovskite compounds and alloys are possibly shown. Temperature behavior of a quantity (T_1) in $SrTiO_3$ are discussed. T_1 is argued to serve as fingerprints in identifying the nature of a local mode. It is found with a dip around 105 K, which defies usual wisdom but proves explicable by both the eight-corner and the rotor model. A mapping may exist between these models. We then model $PbZr_{1-x}Ti_xO_3$ as a lattice of rotors with randomly distributed cubic anisotropy. The x-driven transition is argued to be of percolation type. Consequences are discussed and shown agreeing with observations.

A rotor moves on a sphere or a circle, whose quantum mechanical descriptions appeared shortly after the birth of quantum theory [1]. Elementary rotor does not exist in nature, but it could arise as effective low energy models for a lot of systems [2] such as linear diatomic molecules [3], Josephson array [4] and bosons in an optical lattice [5] to name a few. It has been a prototype model for proof-of-concept elucidation of quantum phase transitions [2,6,7] and was shown to share the same universality class as the antiferromagnetic Heisenberg spin models [8]. In this paper, we present evidence that the behavior of the local mode (as defined in Ref. [9]) of a typical incipient ferroelectric, $SrTiO_3$ (STO), resemble those of an imperfect rotor. This mode roughly represents the motions of the so-called central atom (the Ti atom in STO) and it is closely related to the soft optical phonon that is responsible for ferroelectric instability. In light of this, we discuss the nature of an interesting composition-driven transition in $PbZr_{1-x}Ti_xO_3$ (PZT). We conjecture that this transition may be of percolation type. The implications are also briefly discussed.

The static dielectric constant $\epsilon(T)$ (T being the temperature) of STO obeys the Curie–Weiss' law at high temperatures while saturates at low temperatures, showing no divergence anywhere [10]. The saturation is usually attributed to quantum fluctuations [11,12] and the oxygen octahedra tilting occurring around $T_a = 105$ K [13]. From the mean-field point of view, the temperature dependence of ϵ enters primarily through the bare susceptibility [14], χ_0, which defines the electrical response of the local mode $\vec{u}$ in vanishing molecular field. Different modeling of the local mode leads to different $\chi_0(T)$. In many cases, it is approximated as a bistable system that is characterized by a frequency $k_B T_1/\hbar$, at which the central atom rustles. In pseudo-spin model, T_1 is a constant and measures how fast the spin flips [15], whereas in double-well model, T_1 decreases with decreasing temperature and eventually saturates at low temperatures [16]. A very simple relation exists in the former model, connecting χ_0 to T_1 [17]:

$$\chi_0^{-1} = R \cdot \frac{\tilde{T}_1}{2} \coth\left(\frac{\tilde{T}_1}{2\tilde{T}}\right), \quad (T_1, T) = \frac{\varepsilon_0}{k_B} \cdot (\tilde{T}_1, \tilde{T}) \tag{1}$$

where $R{\sim}3.5$ (see Fig. 1) is a parameter and ε_0 represents an energy unit [18]. This relation is expected to bear a much wider validity beyond a particular model, as suggested in the work of Barret [19]. We will therefore utilize this relation as a definition of T_1. Note that $\chi_0^{-1} \propto T_1$ for $T/T_1 \ll 1$, indicating that T_1 actually measures the low-temperature stiffness of the local mode.

Hence, the temperature dependence of T_1 can serve as fingerprints in identifying the character of the local mode. A detailed understanding of its dynamics may help comprehending the phenemenogy in materials like relaxors, PZT and bismuth ferrite [20–23]. Especially, the subtle effects from the oxygen octahedra tilting may be unraveled by studying T_1 [18], since the local mode must be interacting with this octahedra. Such tilting was argued to play important role in a number of observations [24,25].

Conventional wisdom presumes purely or nearly constant T_1 in STO [10,26,27], producing ostensible agreement with observations [28], which have concealed the rich behavior of T_1 until recently. In Ref. [18], the T_1 was shown with a broad dip around T_a. Below T_a, T_1 increases and eventually saturates as the temperature approaches zero, a behavior defying explanations by both pseudo-spin and double-well models [16]. In what follows, we demonstrate that, such behavior can be explained by the eight-

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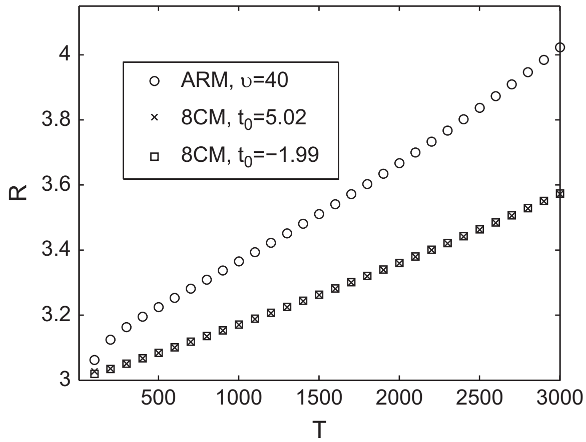

Fig. 1. R as a function of the reduced temperature $\tilde{T}$. It displays rather linear behavior, with very small slope.

corner model (8CM, to be explained below) with a single parameter t_0 representing the inter-corner tunneling energy. Although the observations can be fitted by choosing $t_0 \approx 5.02$, a better fit could be achieved with $t_0 \approx -1.99$. As we will argue, a genuine 8CM excludes negative t_0. Therefore, the fitting with $t_0 \approx -1.99$ must be deciphered otherwise. We reason that, this negative-t_0 case captures essentially the low energy behavior of a rotor model in the presence of weak anisotropy (ARM, to be explained below). In view of this, we may depict the local mode (or more explicitly, the Ti atom) as a rotor. STO may then be regarded as a lattice of rotors and PZT may be modeled as a disordered rotor system. The composition-driven transition is surmised to be of percolation type.

In the 8CM, the local mode occupies the corner states, $|\vec{r}=(X,Y,Z)\rangle$, where $X,Y,Z=\pm 1$, which are defined as the eigenstates of a unit orientation vector $\vec{n}=(n_x,n_y,n_z)$. The 8CM Hamiltonian can be written as

$$\frac{H_{8\mathrm{CM}}}{\varepsilon_0}=\sum_i\sum_{I,J\neq I}t_{I,J}(|I,i\rangle\langle J,i|+|J,i\rangle\langle I,i|),$$

where we use capital letters to tag the corner states while reserving small ones for labeling unit cells. The $t_{I,J}=t(|\vec{r}_I-\vec{r}_J|=r_{IJ})$ signifies the tunneling energy between the corners I and J and ε_0 is the energy unit that will be used as a fitting parameter. Throughout this paper we retain only $t_0 \equiv t(2)$. Two groups of states (each constituting a singlet and a triplet) spaced by $\Delta_0=4t_0$ befall the level structure of this model, as depicted in Fig. 2. For positive t_0, the singlet resides higher than the triplet in energy. The structure is reversed for negative t_0. The χ_0 corresponding to $H_{8\mathrm{CM}}$ can be computed easily and this χ_0 can then be related to T_1 via Eq.(1).

The results are presented in Fig. 3, where the experimentally extracted values of T_1 are represented as black dots. The original data were taken from Ref. [10]. We plot the $(1/\epsilon)$–T curve and identify the slope of its straight section as the inverse Curie constant while its intercept with the T axis as the Curie temperature. With the help of Barret formula [19], T_1 can be deduced [18]. The red curves give the T_1 via Eq.(1), with χ_0 computed according to the 8CM. The solid one corresponds to $t_0 \approx 5.02$, with $\varepsilon_0=5.36$ K. A good fit seems to be found. These parameters lead to a quantum tunneling energy about 27 K, which differs greatly from T_1. As aforementioned, T_1 roughly measures how stiff the system should be in response to external stimuli and therefore reflects the integration of both quantum and thermal fluctuations. On the other hand, an overall better fitting can be obtained if we choose $t_0 \approx -1.99$, with $\varepsilon_0=13$ K, as suggested by the dotted curve. The

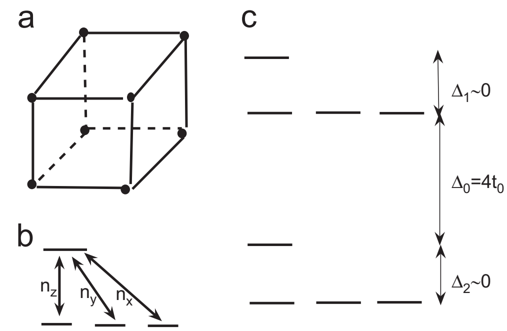

Fig. 2. Illustration of the 8CM. (a) corner states; (b) matrix elements of $\vec{n}$ within the lower singlet–triplet subspace; (c) energy level scheme.

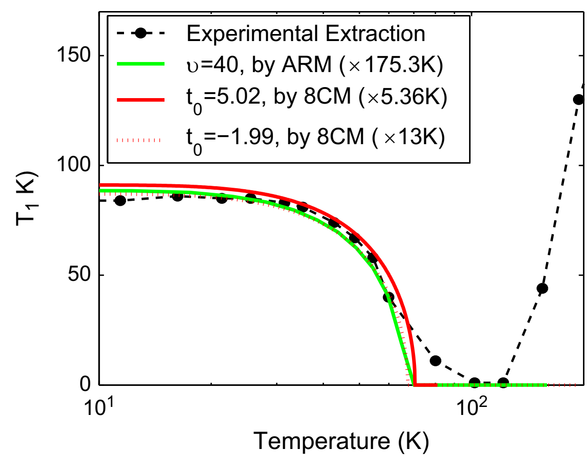

Fig. 3. Experiment vs. theory. Black dots: T_1 deduced as in Ref. [18]. Red solid curve: calculations at t_0=5.02 with the 8CM. Red dotted curve: calculations at $t_0=-1.99$ with the 8CM. Green solid curve: calculations at $\upsilon=40$ with the ARM. Note that, the latter two curves almost coincide. Respective ε_0 s are placed in the parentheses. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this article.)

problem here is that, such negative t_0 cannot exist in the genuine 8CM. Nonetheless, as to be reasoned, the 8CM with $t_0<0$ may effect the low energy physics of a rotor model with weak anisotropy, whereas the strong anisotropy case can be captured by the positive-t_0 8CM. Thus, a correspondence is expected to exist between the ARM and the 8CM. This might be useful in the study of phase transitions in rotor systems.

The conventional rotor model contains just the kinetic energy, which in the dimensionless form is written as

$$H_0=\frac{\hat{\Omega}^2}{2},$$

with $\hat{\Omega}$ being the angular momentum operator of the rotor (the central atom). Now add some anisotropy to it and the ARM results. This anisotropy is assumed observing cubic symmetry and we put it as

$$H'=\upsilon(n_x^4+n_y^4+n_z^4),$$

where υ denotes the strength of the anisotropy. We note that, the ARM ensues directly from Eq. (1) in Ref. [29] or can be derived from the effective model in Ref. [9] if one freezes the magnitude of the order parameter therein. H' features eight (six) minima on the sphere in the directions like (1,1,1) ((0,0,1)) for positive (negative) υ. Imagine that υ is sufficiently large so that H' dominates the low energy states. On such occasions, the rotor should primarily dwell around the minima, among which tunneling could be induced by H_0. Hence, one should revisit the conventional 8CM for positive υ.

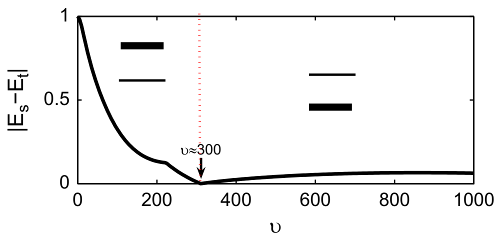

Fig. 4. The singlet–triplet structure (of the ARM) turns upside down when tuning υ. E_s: singlet energy; E_t: triplet energy. The reversion happens around 300. Inset thick (thin) line stands for the triplet (singlet). The kink in between 200 and 300 marks a discontinuity in $\partial|E_s-E_t|/\partial\upsilon$, implying a transition.

To demonstrate this, we look at the energy level structure. The ARM has been solved for various υ ranging from zero to as large as 1000 [30]. Interestingly, despite the values of υ, the low energy ARM level structure always resembles that of the 8CM, i.e., consisting of a singlet and a triplet. At vanishing υ, the singlet is simply the s orbital while the p orbital make up the triplet, with the latter on top of the former. As displayed in Fig. 4, this trait persists all the way up to υ~300, at which point the structure is reversed. Making connections to the 8CM, one may say that t_0 changes sign across this point. The conventional 8CM corresponds to the large υ extreme [31], and therefore should have positive t_0. Allowing negative t_0 and considering that the orientation vector has the same contents in both models, the 8CM can then be taken as a good approximation of the ARM with any υ. A deeper relation between the 8CM and rotor model is revealed by symmetry considerations: there is an approximate continuous rotational symmetry that is respected by the former. Actually, omitting the upper singlet–triplet group shown in Fig. 2(c), which is possible in studying low energy physics if Δ_0 is sufficiently large, one arrives at $H_{8\mathrm{cm}}/\varepsilon_0 \approx (\Delta_2/2)(n_x^2+n_y^2+n_z^2)$. As the H_0, it respects the $O(3)$ continuousrotational symmetry, a situation analogous to what is considered in Ref. [32].

With the above mapping, we expect that the $t_0 \approx -1.99$ in the 8CM should represent a small υ in the ARM. To confirm this, we have fitted the T_1 using a χ_0 that is computed with the ARM. As we see in Fig. 3, the green curve obtained by ARM with $\upsilon\approx40$ nicely overlaps with the red dotted one based on 8CM with negative t_0. In this view, we propose that the local mode of STO at low temperatures might realize a lattice of rotors subjected to weak cubic anisotropy.

We proceed to discuss possible implications for PZT, a pseudo-cubic solid solutions characterized by site and charge disorder. In its phase diagram, a T–x line separates the paraelectric phase from the ferroelectric one [33,34]. Within the ferroelectric region, several orders were found. For $x<0.48$, the compound comes in the rhombohedral phase, while it transits to the tetragonal phase when x is tuned beyond 0.50 [33,34]. In between, a monoclinic phase, whose polarization is confined to symmetry planes rather than symmetry lines, might exist. This intermediate phase was proposed to be a salient feature of materials showing giant piezoelectric response [35,36]. However, its nature remains yet to be clarified. On one hand, this phase was argued to result from long-range polarization rotation [33,34] whereas on the other hand, recent work challenges this picture by proving that, no such rotation happens locally at all [37]. Both first principles [34] and Ginzberg–Landau–Devonshire-type theory [38] have been performed on this issue and vindicated the existence of the monoclinic phase, but it is desirable to gain further insights.

The first step might be to construct a tractable, atomistic and physically meaningful model. To this end, we notice that the ground state of the ARM favors diagonal order and therefore the rhombohedral phase for positive υ while axial order and therefore the tetragonal phase for negative υ [29]. Assuming that the validity of ARM and inspecting on the evolution of PZT as x is varied, one may conjecture that, the value of υ is essentially determined by the presence of which element, Ti or Zr; namely $\upsilon_{\mathrm{Ti}}>0$ and $\upsilon_{\mathrm{Zr}}<0$. Counting on these, we naturally arrive at a model Hamiltonian as follows:

$$H_{\mathrm{PZT}} = \sum_i \left\{ \frac{\hat{\Omega}_i^2}{2} + \upsilon_i (n_{ix}^4 + n_{iy}^4 + n_{iz}^4) \right\} + \sum_{i,j;\mu\nu} K_{ij}^{\mu\nu} n_{i\mu} n_{j\nu} + V,$$

where the Latin subscripts label unit cells of PZT and Greek ones label the components of the orientation vectors and V includes the terms involving strain. The disorder effects are entertained by υ_i (and perhaps the rotation inertia that is also random but neglected here), which is supposed to depend on which element takes up the i-th unit cell. At $x=0$ and $x=1$, the system is uniform but in the rhombohedral and tetragonal phases, respectively. A rhombohedral–tetragonal transition is expected to occur somewhere. It might befall abruptly or through gradual polarization rotation. Note that H_{PZT} is similar to 3D random Ising model, which has been utilized to understand PZT [39].

We for the moment switch off the $K_{ij}^{\mu\nu}$ for $\upsilon_i \cdot \upsilon_j < 0$, because: (1) the interactions seem extremely axial [29], i.e., $K_{ij}^{\mu\nu}$ is maximum for $\mu=\nu$ and $\vec{R}_{ij} \propto e_\mu$, where $\vec{R}_{ij}$ joins the i-th to the j-th unit cell and e_μ denotes the axial direction; (2) a rotor tends to point along the diagonal if $\upsilon>0$ while along the axis if $\upsilon<0$, and therefore, two rotors with opposite signs of υ cannot efficiently align to maximize their interactions because of (1). Under this simplification, the rotors are anticipated to aggregate into two types of disconnected clusters with diagonal and axial order, respectively. At $x=0$, there is no axial cluster. Increasing x a bit, axial clusters are expected to form and be scattered in the background with diagonal order, like lakes in a continent, but the system on the whole is still in the rhombohedral phase. Increasing x further, the axial clusters should grow in size and eventually have a significant probability to link all together to reach full connectivity at a critical point x_c. Simultaneously, the continent is expected to become boggled and look like islands poping out in the sea. Eventually, the system should transit to the tetragonal phase as x crosses x_c. Conspicuously, we are describing an ideal percolation transition [2,40] that is expected to take place when neither the sea nor the continent prevails. Two things can be immediately inferred. Firstly, $x_c\approx0.5$, assuming that the free energy of an axial cluster is the same as that of a diagonal one of the same size. This value is larger than the one (≈0.3) for the usual percolation transition in a simple cubic lattice [41], maybe because of the interaction between the atoms within a cluster. Secondly, the phase boundary should be vertical as is usually so with percolation transitions. Both features agree with the phase diagram of PZT. Nonetheless, the ideal transition described here seemingly does not allow intermediate phases such as the monoclinic phase. Perhaps, it has to be modified when the coupling is turned on between the two types of clusters. Detailed investigations along this line need be conducted in the future.

To summarize, we have discussed the temperature behavior of a quantity named T_1, which is construed as a measure of the stiffness of the local mode of STO in dielectric response. The experimentally extracted T_1 was found to exhibit a dip, which cannot reconcile with the most often used models. However, this dip can be reproduced by the 8CM with either a positive or a negative tunneling integral, the latter being overally better. In the meanwhile, such behavior can be equally well explained by a rotor model in the presence of weak anisotropy. By examining the low energy level structure and the symmetry, we argue that a mapping could exist between the 8CM and the ARM. This finding indicates that these two models might belong to the same universality class and thus share the same long-wavelength behavior. We further extend the discussions to PZT. A microscopic model is hypothesized to understand the quantum phase transition driven by the Ti concentration. This model essentially describes a lattice of rotors subject to cubic anisotropies whose signs are randomly distributed according to a bimodal law. The composition-driven transition is then proposed to be a percolation transition between the diagonal order and the axial order. Consequently, we expect an x_c of around 0.5 and a vertical phase boundary, in agreement with the observations. Nevertheless, the monoclinic phase seems missing in this simple picture. Hopefully, this can be resolved by inclusion of some couplings between rotors with opposite signs of υ. Our work may stimulate further theoretical and experimental work.

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